Weekly questions (without solution)

Week 1 introduction

1. What is a decision tree?
2. What is rule-based?
3. What is greedy approach in decision tree?
4. What is overfitting? How can we overfit? How can we avoid overfitting in decision trees?
5. What is supervised learning?
6. What is unsupervised learning?
7. What is reinforcement learning?

Week 2 optimization and search

1. What is search landscape? What is the z-axis?
2. What is the goal of optimalization algorithms?
3. What is the difference between global and local optima?
4. What is the difference between continuous and discrete optimization? Name some discrete optimalization methods and name a continuous optimalization method.
5. What is exhaustive search? When is it beneficial to use es (what type of graph)? Is it guaranteed to find the best solution?
6. What is greedy search?
7. What is hill climbing? When do we stop?
8. What is exploitation and exploration? What category is exhaustive search and hill climbing?
9. What is simulated annealing?
10. What is gradient ascent/descent?
11. Gradient ascent and hill climbing are quite similar, and they are based almost exclusively on exploitation. Can you think of any additions to these algorithms in order to do more exploration?